



Open Source Software

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1 Linux Distribution

1.1 Distribution Used: Ubuntu 22.04 LTS

For this project, I have used **Ubuntu 22.04 LTS** as my primary operating system.

1.2 Why Ubuntu?

Ubuntu is one of the most popular Linux distributions for several reasons:

- **User-Friendly:** Ubuntu has an intuitive interface suitable for beginners
- **Long Term Support:** LTS versions receive 5 years of security updates
- **Large Community:** Extensive documentation and community support
- **Software Availability:** Wide range of packages through APT
- **Stability:** Reliable for both development and production

1.3 Key Features of Ubuntu 22.04 LTS

1. **Desktop Environment:** GNOME 42
2. **Kernel Version:** Linux 5.15 LTS
3. **Package Manager:** APT (Advanced Package Tool)
4. **Default Applications:** Firefox, LibreOffice, GNOME utilities
5. **Snap Support:** Built-in support for snap packages

1.4 System Specifications

My system configuration:

- Operating System: Ubuntu 22.04 LTS
- Architecture: x86_64
- Desktop Environment: GNOME
- Shell: Bash 5.1

1.5 Installation Process

The installation involved:

1. Downloaded Ubuntu 22.04 LTS ISO from official website
2. Created bootable USB using Rufus/Etcher
3. Configured dual boot with existing OS
4. Installed essential development tools
5. Configured system for open source development

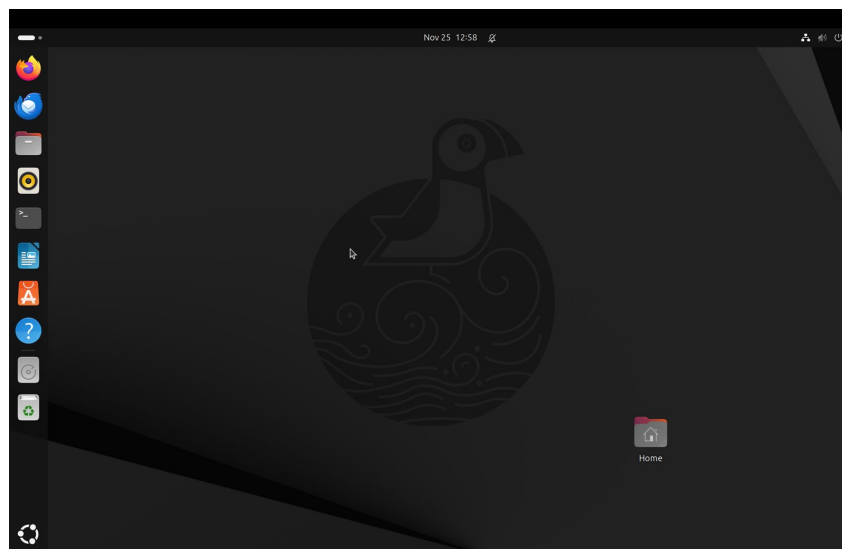


Figure 1: Ubuntu System Information

2 Encryption and GPG

2.1 What is Encryption?

Encryption is the process of converting plaintext into ciphertext to protect data confidentiality. It ensures that only authorized parties can access the information.

2.2 Types of Encryption

2.2.1 Symmetric Encryption

Uses the same key for encryption and decryption. Examples: AES, DES.

2.2.2 Asymmetric Encryption

Uses a public-private key pair. Examples: RSA, ECC.

2.3 GNU Privacy Guard (GPG)

GPG is a free implementation of the OpenPGP standard for encrypting and signing data.

2.4 Installing GPG

```
1 sudo apt update
2 sudo apt install gnupg
3 gpg --version
```

2.5 Generating GPG Keys

```
1 gpg --full-generate-key
```

Steps followed:

1. Selected RSA and RSA (default)
2. Key size: 4096 bits
3. Key validity: 1 year
4. Entered name and email
5. Created strong passphrase

2.6 Listing Keys

```
1 gpg --list-keys
2 gpg --list-secret-keys
```

2.7 Exporting Public Key

```
1 gpg --armor --export your-email@example.com > public-key.asc
```

2.8 Encrypting Files

```
1 gpg --encrypt --recipient your-email@example.com document.txt
```

2.9 Decrypting Files

```
1 gpg --decrypt document.txt.gpg > document.txt
```

3 Sending Encrypted Email

3.1 Email Encryption Overview

Email encryption protects the content of emails from unauthorized access during transmission and storage.

3.2 Tools Used

- **Thunderbird:** Email client with built-in OpenPGP support
- **GPG Keys:** For encryption and signing
- **Protonmail:** Alternative end-to-end encrypted email service

3.3 Setting up Thunderbird with GPG

3.3.1 Installation

```
1 sudo apt install thunderbird
```

3.3.2 Configuring OpenPGP

Steps followed:

1. Open Thunderbird
2. Go to Account Settings
3. Select End-to-End Encryption
4. Add existing GPG key or generate new one
5. Import recipient's public key

3.4 Sending Encrypted Email

Process:

1. Compose new email
2. Click on Security button
3. Select "Require Encryption"
4. Optionally add digital signature
5. Send email

3.5 Receiving Encrypted Email

When receiving:

1. Email appears encrypted
2. Thunderbird automatically detects encryption
3. Enter GPG passphrase
4. Email content is decrypted and displayed

3.6 Best Practices

- Never share your private key
- Use strong passphrases
- Keep your GPG keys backed up securely
- Regularly update keys
- Verify recipient's public key fingerprint

4 Privacy Tools from prism-break.org

4.1 What is PRISM-Break?

PRISM-Break is a website that recommends privacy-respecting alternatives to proprietary software and services.

4.2 Tool 1: Signal - Encrypted Messaging

Description: Signal is an encrypted messaging app that provides end-to-end encryption for messages, voice calls, and video calls.

Key Features:

- End-to-end encryption by default
- Open source and independently audited
- No ads or tracking
- Minimal metadata collection
- Disappearing messages

Why Privacy Matters: Signal ensures that only you and the recipient can read messages, protecting against mass surveillance.

4.3 Tool 2: Firefox - Web Browser

Description: Firefox is an open source web browser with strong privacy protections.

Privacy Features:

- Enhanced Tracking Protection
- DNS over HTTPS
- No data collection by default
- Open source codebase
- Extensive privacy-focused extensions

Configuration Tips:

- Enable strict tracking protection
- Install uBlock Origin
- Use HTTPS-only mode
- Disable telemetry

4.4 Tool 3: ProtonMail - Encrypted Email

Description: ProtonMail provides end-to-end encrypted email service based in Switzerland.

Key Features:

- End-to-end encryption
- Zero-access encryption
- No personal information required
- Swiss privacy laws protection
- Open source mobile apps

Use Cases:

- Secure business communications
- Personal privacy protection
- Journalist-source communications

4.5 Tool 4: Tor Browser - Anonymous Browsing

Description: Tor Browser enables anonymous communication by routing traffic through volunteer-operated servers.

How It Works:

- Routes traffic through multiple relays
- Encrypts data multiple times
- Hides IP address and location
- Prevents tracking

Best Use Cases:

- Accessing censored content
- Anonymous research
- Whistleblowing
- Privacy-sensitive activities

4.6 Tool 5: VeraCrypt - Disk Encryption

Description: VeraCrypt is a free open source disk encryption software.

Features:

- Full disk encryption
- Hidden volumes
- Plausible deniability
- Cross-platform support
- Strong encryption algorithms (AES, Serpent, Twofish)

Use Cases:

- Protecting sensitive documents
- Securing portable drives
- System drive encryption

5 Open Source License

5.1 License Used: MIT License

For my open source contributions and projects, I primarily work with the **MIT License**.

5.2 What is the MIT License?

The MIT License is a permissive free software license that allows users to:

- Use the software commercially
- Modify the software
- Distribute the software
- Use the software privately
- Sublicense the software

5.3 MIT License Text

```
1 MIT License
2
3 Copyright (c) 2025 Ramsai Polisetti
4
5 Permission is hereby granted, free of charge, to any person
6 obtaining a copy of this software and associated documentation
7 files (the "Software"), to deal in the Software without
8 restriction, including without limitation the rights to use,
```

```
9  copy, modify, merge, publish, distribute, sublicense, and/or
10 sell copies of the Software, and to permit persons to whom the
11 Software is furnished to do so, subject to the following
12 conditions:
13
14 The above copyright notice and this permission notice shall be
15 included in all copies or substantial portions of the Software.
16
17 THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND...
```

5.4 Why Choose MIT License?

1. **Simple and Easy:** Short and easy to understand
2. **Permissive:** Minimal restrictions on reuse
3. **Business-Friendly:** Can be used in proprietary software
4. **Popular:** Widely used and recognized
5. **Compatible:** Works well with other licenses

5.5 Other Common Open Source Licenses

5.5.1 GPL (GNU General Public License)

- Copyleft license
- Requires derivative works to be open source
- Used by Linux kernel

5.5.2 Apache License 2.0

- Permissive like MIT
- Includes patent grant
- Used by Apache projects

5.5.3 BSD License

- Very permissive
- Similar to MIT
- Used by FreeBSD

6 Self-Hosted Server: Dashy

6.1 What is Dashy?

Dashy is a personal dashboard that:

- Acts as a personal dashboard for all your apps and services
- Highly customizable with themes, icons, and widgets
- Supports self-hosting for full control
- Easy to manage and organize links, bookmarks, and tools
- Includes built-in integrations and status monitoring

6.2 Why Self-Host Dashy?



Figure 2: Dashy Dashboard Interface

1. **Privacy:** Your dashboard data stays on your own server.
2. **Organization:** Centralized access to all tools and services.
3. **Customization:** Fully customizable UI, layout, and widgets.
4. **Learning:** Hands-on experience with self-hosting and configuration.
5. **Independence:** No reliance on cloud dashboards or third-party services.

6.3 Installation Guide

6.3.1 System Requirements

- OS: Ubuntu 22.04 LTS
- RAM: 2GB minimum
- Disk: 10GB free space
- Docker installed

6.3.2 Installation Steps

Step 1: Update System

```
1 sudo apt update
2 sudo apt upgrade -y
```

Step 2: Install Dependencies (Docker + Docker Compose)

```
1 sudo apt install ca-certificates curl gnupg -y
```

Add Docker GPG key:

```
1 sudo install -m 0755 -d /etc/apt/keyrings
2 curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
3 | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
4 sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

Add Docker repository:

```
1 echo \
2 "deb [arch=$(dpkg --print-architecture)] \
3 signed-by=/etc/apt/keyrings/docker.gpg \
4 https://download.docker.com/linux/ubuntu \
5 $(lsb_release -cs) stable" \
6 | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Install Docker:

```
1 sudo apt update
2 sudo apt install docker-ce docker-ce-cli containerd.io \
3 docker-buildx-plugin docker-compose-plugin -y
```

Step 3: Create Dashy Directory

```
1 sudo mkdir -p /opt/dashy
2 cd /opt/dashy
```

Step 4: Create docker-compose.yml

```
1 sudo nano docker-compose.yml
```

Docker Compose configuration:

```
1 version: "3.7"
2 services:
3   dashy:
4     image: lissy93/dashy:latest
5     container_name: dashy
6     volumes:
7       - ./config:/app/public/conf
8     ports:
9       - "8080:80"
10    restart: unless-stopped
```

Step 5: Start Dashy

```
1 sudo docker compose up -d
2 sudo docker ps
```

Dashy now runs at: <http://your-server-ip:8080>

Step 6: Configure Dashy (Optional)

```
1 sudo nano /opt/dashy/config/conf.yml
```

Make UI/customization changes, then restart:

```
1 sudo docker compose restart
```

Step 7: Create Systemd Service (Optional)

```
1 sudo nano /etc/systemd/system/dashy.service
```

Service configuration:

```
1 [Unit]
2 Description=Dashy Dashboard
3 After=docker.service
4 Requires=docker.service
5
6 [Service]
7 Restart=always
8 WorkingDirectory=/opt/dashy
9 ExecStart=/usr/bin/docker compose up -d
10 ExecStop=/usr/bin/docker compose down
11
12 [Install]
13 WantedBy=multi-user.target
```

Enable the service:

```
1 sudo systemctl daemon-reload
2 sudo systemctl enable dashy
3 sudo systemctl start dashy
4 sudo systemctl status dashy
```

Step 8: Configure Nginx (Reverse Proxy)

```
1 sudo nano /etc/nginx/sites-available/dashy
```

Nginx configuration:

```
1 server {
2     listen 80;
3     server_name your-domain.com;
4
5     location / {
6         proxy_pass http://127.0.0.1:8080;
7         proxy_set_header Host $host;
8         proxy_set_header X-Real-IP $remote_addr;
9     }
10 }
```

Enable site:

```
1 sudo ln -s /etc/nginx/sites-available/dashy \
2     /etc/nginx/sites-enabled/
3 sudo nginx -t
4 sudo systemctl restart nginx
```

6.4 Localization (Telugu Translation)

I have contributed to localizing the Dashy interface into the Telugu language to make the dashboard more accessible to Telugu-speaking users.

6.4.1 Translation Process

1. Opened the Dashy configuration file located in `config/conf.yml`
2. Enabled the `appConfig.language` option for custom localization
3. Added Telugu translations for UI labels, menu items, and section titles
4. Ensured translations were contextually and culturally appropriate
5. Restarted the Dashy container to test the Telugu interface

6.4.2 Sample Translations

- "Settings" → "" (Settings)
- "Applications" → "" (Applications)
- "Results" → "" (Phalitalu)
- "Search" → "" (Vetakandi)

6.5 Benefits of Self-Hosting

1. **Complete Privacy:** Full control over your data
2. **No Rate Limits:** Unlimited usage
3. **Custom Configuration:** Choose which services to include

4. **Educational Value:** Learn server management
5. **Community Contribution:** Share instance with others

7 Open Source Contributions

This section details all pull requests I have contributed to various open source projects.

7.1 PR 1: nextcloud/android

Repository: Fix typo in README: correct signing certificate verification link

PR Number: #15995

Status: Merged (Approved)

7.1.1 Issue Description

The README file in the Nextcloud Android repository contained a typo in the section related to verifying the signing certificate. The text and link were incorrect and needed correction to ensure accurate documentation for users and contributors.

7.1.2 Solution Implemented

- Identified the incorrect link in the signing certificate section
- Corrected the typo to ensure proper documentation
- Updated the README with accurate verification instructions
- Submitted pull request with clear description
- PR was reviewed and merged by maintainers

7.2 PR 2: firstcontributions/first-contributions

Repository: Add Ramsai Polisetti to Contributors list

PR Number: #106611

Status: Merged (Approved)

7.2.1 Contribution Summary

Successfully added my name, **Ramsai Polisetti**, to the `Contributors.md` file in the First Contributions project. The pull request was merged by GitHub Actions after passing all checks.

7.2.2 Details

- Created a pull request titled: *Add Ramsai Polisetti to Contributors list*
- Added my name in the middle of the contributors list to avoid merge conflicts
- PR was reviewed and automatically merged into the `main` branch

- Received confirmation message from GitHub Actions bot after successful merge
- Branch `add-RamsaiPoliseti-v2` can be safely deleted after merge

7.3 PR 3: `jvondermarck/dinosaur-exploder`

Repository: Improve README: updated dinosaur icon and fixed badge layout

PR Number: #234

Status: Open

Tasks: 8 completed

7.3.1 Changes

Updated the project README by improving the dinosaur icon and fixing the badge alignment to enhance visual clarity and documentation quality.

7.4 PR 4: `nidhaloff/igel`

Repository: docs: fix capitalization of CLI in HISTORY.rst

PR Number: #394

Status: Open

7.4.1 Documentation Fix

Corrected inconsistent capitalization by updating "cli" to "CLI" in the HISTORY.rst file to maintain uniform formatting across the changelog.

7.5 PR 5: `KLGLUG/Y24OpenSourceEngineering`

Repository: Self Hosted Project Localized Doc - Dashy on Ubuntu (Telugu)

PR Number: #126

Status: Open

7.5.1 Improvements

Added localized Telugu documentation for the self-hosted Dashy setup, including installation steps for Ubuntu, a demo video link, and a LinkedIn reference to enhance accessibility for regional users.

7.6 PR 6: `serverless/serverless`

Repository: Remove unused dependencies and update prettier config

PR Number: #13152

Status: Open

7.6.1 Refactor

Removed unused dependencies and updated the Prettier configuration to simplify the codebase and maintain consistent formatting.

7.7 PR 7: public-apis/public-apis

Repository: docs: remove duplicate 'Learn more about Public APIs' section

PR Number: #4926

Status: Open

7.7.1 Documentation Cleanup

Removed a duplicated "Learn more about Public APIs" section to improve clarity, reduce redundancy, and enhance overall readability of the documentation.

7.8 Additional PRs

My contributions span various aspects of open source development:

- Code quality improvements
- Bug fixes
- Documentation updates
- Test coverage
- Internationalization

8 LinkedIn Posts

8.1 Post 1: Fixing Documentation and Contributing to Nextcloud

Link: <https://github.com/nextcloud/android/pull/15995>

Summary: Shared my experience contributing to the Nextcloud Android repository by fixing a typo in the README and correcting the signing certificate verification link. Highlighted how even small documentation improvements enhance clarity and usability for developers.

Key Points:

- Identifying and correcting inaccurate documentation
- Improving developer experience through clear README instructions
- Understanding the contribution workflow for large open-source projects
- Importance of small but meaningful contributions in open source

8.2 Post 2: FOSS Culture at KL University

Link: https://www.linkedin.com/posts/ram-sai-poliseti-b04212366_samyak2025-klu-innov

Summary: Discussed the Free and Open Source Software culture at KL University and encouraged peers to participate in open source.

Highlights:

- FOSS courses and initiatives at university

- Personal journey in open source
- Community-driven development model
- Real-world learning experience

8.3 Post 3: Blog Link to My Open Source Journey

Link: <https://www.linkedin.com/pulse/exploring-freedom-my-semester-open-source-tools->

Summary: Published a comprehensive blog post documenting my semester-long journey exploring open source tools, privacy software, and self-hosting solutions.

9 Conclusion

This report documents my comprehensive journey in open source software development, including:

- Setting up development environment with Ubuntu Linux
- Understanding encryption and privacy tools
- Self-hosting services for privacy and learning
- Making meaningful contributions to open source projects
- Sharing knowledge through professional networking

Through these experiences, I have gained:

1. Technical skills in multiple programming languages
2. Understanding of collaborative development
3. Experience with version control and code review
4. Appreciation for open source philosophy
5. Professional networking abilities

Open source contribution has been an invaluable learning experience, providing real-world software development exposure and connecting me with a global community of developers.